Free-fall lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Estimate g from measured motion and discuss measurement limits.

SUCCESS CRITERIA - I can...

1. Keep supplied examples separate from measurements.
2. Find acceleration from a velocity-time slope.
3. Use repeated trials and explain uncertainty.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “slope” and “trial” to explain your model or result. Sentence starters are optional.

slope: vertical change divided by horizontal change on a graph

trial: one run or repetition of a test

acceleration: rate of change of velocity

01 Bring back an earlier idea

Two equal 3000 kg masses are 5 m apart.
Use $G=6.67\times10^{-11}$ N·m²/kg².
Find force. By what factor is an answer wrong if r rather than r² is used?

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $a = \Delta v/\Delta t$; $v_f = v_i + a\Delta t$.
Velocity-time slope estimates acceleration. Downward is positive. Use the approved apparatus; distinguish supplied from measured data.

Worked reasoning: Supplied trial values 8, 10, 12 m/s² have mean 10 m/s² and range 8-12. The spread describes variation; averaging does not remove every systematic error.

Your turn: Plot the three values on a number line and mark the mean.
Name an error that could shift all trials together.

Sentence starter: The step I would change is … . The evidence is … .

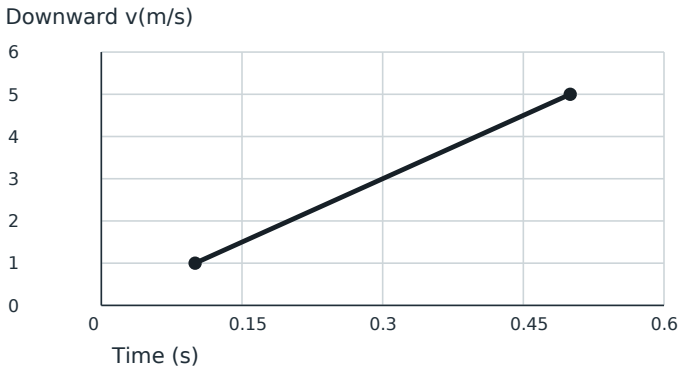
Marzano check - circle: ① ② ③

03 Represent and interpret

Use the supplied velocity-time graph to estimate acceleration.
Show the change in velocity and elapsed time.

Word help: estimate = find an approximate value

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

04 Solve and justify

Trial estimates are 9.6, 9.9, 10.2 m/s².
Find mean, half-range and percent deviation of mean from 9.8 m/s².

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Plan three repeated trials.
Name a controlled condition, a measurement you will record and a reason not to include a badly pushed release without comment.

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: ① ② ③

06 Choose a method

A 4 kg lamp hangs at rest from a vertical rope.
Use g=9.8.
Find tension. Then find tension if it accelerates upward at 3 m/s².

Word help: tension = the pulling force through a stretched rope or cable

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

07 Exit check

A graph fit uses v=.15 m/s at .02 s and v=2.11 m/s at .22 s.
Find g.
Explain why fitting a good interval can be better than one noisy instantaneous value.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: ① ② ③

08 STRETCH: Apply it another way

Supplied estimates are 9.4, 9.8, 10.2 m/s².
Find mean, half-range and percent deviation of mean from 9.81. Could zero spread guarantee accuracy?

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③